

CHAPTER 4. ADDITIONAL APPLICATIONS OF THE DERIVATIVE

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4.1 Extreme Values of a Continuous Function

Definition 1 (Extreme values)

Let c be a number in the domain D of a function f . Then $f(c)$ is the

- **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D .
- **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D .

- An absolute maximum or minimum is sometimes called a **global** maximum or minimum.
- The maximum and minimum values of f are called **extreme values** or **absolute extrema** of f .

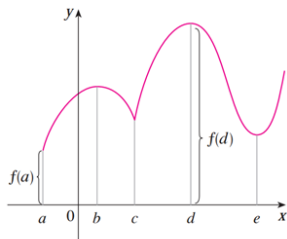


FIGURE 2

Abs min $f(a)$, abs max $f(d)$,
loc min $f(c)$, $f(e)$, loc max $f(b)$, $f(d)$

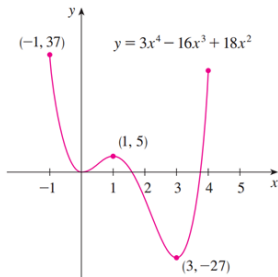
Definition 2 (Local maximum and minimum values)

The number $f(c)$ is a

- **local maximum** value of f on D if $f(c) \geq f(x)$ when x is near c .
- **local minimum** value of f on D if $f(c) \leq f(x)$ when x is near c .

The graph of the function

$$f(x) = 3x^4 - 16x^3 + 18x^2, \quad -1 \leq x \leq 4$$



Theorem 3 (The Extreme Value Theorem)

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

How to find these extreme values:

1. We start by looking for local extreme values.
2. And then the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

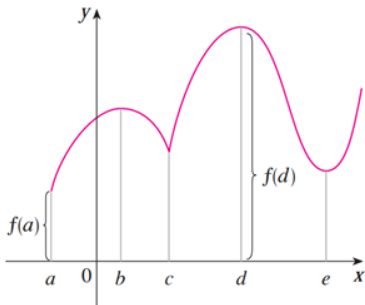
Looking for local extreme values

Theorem 4 (Critical number theorem)

If f has a local maximum or minimum at c , then c is a critical number of f .

Definition 5

A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.



Example 6

Find the critical numbers for the given functions

a. $f(x) = x^3 - 6x^2 + 9x + 10$ **b.** $f(x) = \frac{e^x}{x-2}$ **c.** $f(x) = 2\sqrt{x}(6-x)$

Example 7

Find the critical numbers for $f(x) = |x|$ on $[-1, 1]$

The Closed Interval Method

Extreme values of a continuous function f on an interval $[a, b]$

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Example 8

Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1, \quad -1 \leq x \leq 4$$

Example 9

Find the absolute extrema of the function

$$g(x) = x^{2/3}(5 - 2x), \quad -1 \leq x \leq 2.$$

Example 10

Find the absolute maximum and minimum values of the function

$$f(t) = 2 \cos t + \sin 2t, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Example 11

The Hubble Space Telescope was deployed on April 24, 1990, by the space shuttle *Discovery*. A model for the velocity of the shuttle during this mission, from liftoff at $t = 0$ until the solid rocket boosters were jettisoned at $t = 126$ s, is given by

$$v(t) = 0.001302t^3 - 0.09029t^2 + 23.61t - 3.083$$

(in feet per second). Using this model, estimate the absolute maximum and minimum values of the *acceleration* of the shuttle between liftoff and the jettisoning of the boosters.



Example 12

Show that if a rectangle with fixed perimeter P is to enclose the largest area, it must be a square.

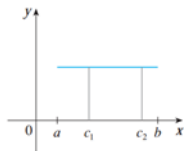
4.2 The Mean Value Theorem

Theorem 13 (Rolle's Theorem)

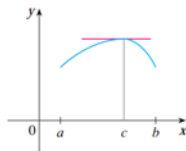
Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$

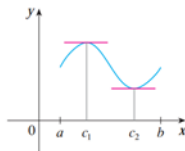
Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.



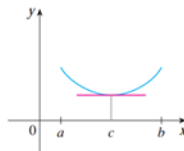
(a)



(b)



(c)



(d)

Example 14

Show that there is a unique solution to the equation $x^3 + x - 1 = 0$ on the interval $(0, 1)$.

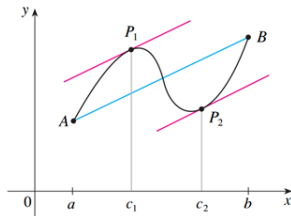
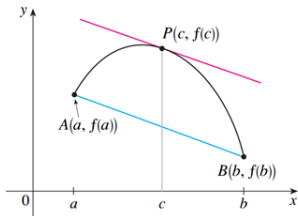
Theorem 15 (The Mean Value Theorem)

Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{or} \quad f(b) - f(a) = f'(c)(b - a) \quad (1)$$



Example 16

Show that the function $f(x) = x^3 - x$ satisfies the hypotheses of the Mean Value Theorem on the closed interval $[0, 2]$ and find a number $c \in (0, 2)$ so that

$$f'(c) = \frac{f(2) - f(0)}{2 - 0}$$

Example 17

If an object moves in a straight line with position function $s = f(t)$, then the average velocity between $t = a$ and $t = b$ is

$$\frac{f(b) - f(a)}{b - a}$$

and the velocity at $t = c$ is $f'(c)$. Thus the Mean Value Theorem tells us that at some time $t = c$ between a and b the instantaneous velocity $f'(c)$ is equal to that average velocity.

Applications of the Mean Value Theorem

Obtain information about a function from information about its derivative

Example 18

Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x . How large can $f(2)$ possibly be?

Applications of the Mean Value Theorem

Zero-derivative Theorem

Theorem 19 (Zero-derivative Theorem)

If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

Corollary 20

If $f'(x) = g'(x)$ for all $x \in (a, b)$, then $f - g$ is constant on (a, b) , that is,

$$f(x) = g(x) + C$$

where C is a constant.

Example 21

Prove the identity $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$.

4.3 Using Derivative to Sketch the Graph of a Function

What Does f' Say About f ?

Increasing/Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.
- (b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

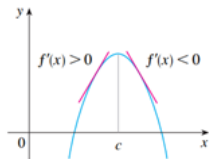
Example 22

Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.

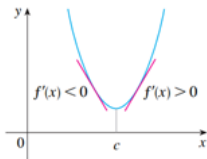
The First Derivative Test

Suppose that c is a critical number of a continuous function f .

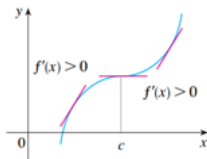
- (a) If f' changes from positive to negative at c , then f has a local maximum at c .
- (b) If f' changes from negative to positive at c , then f has a local minimum at c .
- (c) If f' does not change sign at c (for example, if f' is positive on both sides of c or negative on both sides), then f has no local maximum or minimum at c .



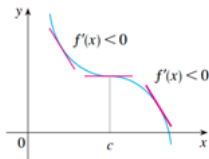
(a) Local maximum



(b) Local minimum



(c) No maximum or minimum



(d) No maximum or minimum

Example 23

Find the local minimum and maximum values of the function

$$f(x) = 3x^4 - 4x^3 - 12x^2 + 5$$

Example 24

Find the local minimum and maximum values of the function

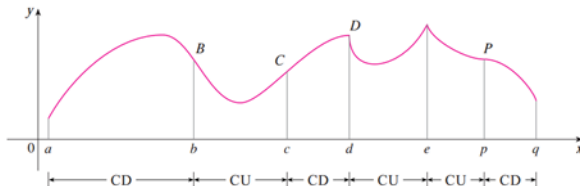
$$g(x) = x + 2 \sin x \quad 0 \leq x \leq 2\pi$$

4.3 Using Derivative to Sketch the Graph of a Function

What Does f'' Say About f ?

Definition 25 (Concavity)

- If the graph of f lies above all of its tangents on an interval I , then it is called **concave upward** on I .
- If the graph of f lies below all of its tangents on I , it is called **concave downward** on I .



Concavity Test

- (a) If $f''(x) > 0$ for all $x \in I$, then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ for all $x \in I$, then the graph of f is concave downward on I .

Example 26

Find where the graph of $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is concave upward and where it is concave downward.

Definition 27 (Inflection Point)

A point P on a curve $y = f(x)$ is called an **inflection point** if it is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

The Second Derivative Test

Suppose that f'' is continuous near c .

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

Example 28

Discuss the curve $y = 3x^4 - 4x^3$ with respect to concavity, points of inflection, and local maxima and minima. Use this information to sketch the curve.

4.4 Curve Sketching with Asymptotes

(i) **Horizontal Asymptotes.** If either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L,$$

then the line $y = L$ is a horizontal asymptote of the curve $y = f(x)$.

(ii) **Vertical Asymptotes.** If at least one of three following limits

$$\lim_{x \rightarrow a} f(x), \quad \lim_{x \rightarrow a^+} f(x), \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x)$$

is infinite, then the line $x = a$ is a vertical asymptote of the curve $y = f(x)$.

(iii) **Slant Asymptotes.** If

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$$

then the line $y = mx + b$ is called a **slant asymptote** of the curve $y = f(x)$.

Example 29

Sketch the graph of $f(x) = \frac{x+1}{x-1}$.

Example 30

Discuss and sketch the graph of $f(x) = \frac{x^2 + x + 2}{x - 1}$.

4.4 Curve Sketching with Asymptotes

Graphing with vertical tangents and cusps

Vertical tangents and cusps

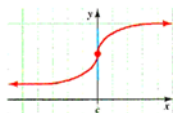
Suppose the function f is continuous at the point $P(c, f(c))$.

- The graph of f has a **vertical tangent** line at P if

$$\lim_{x \rightarrow c^-} f'(x) \text{ and } \lim_{x \rightarrow c^+} f'(x) \text{ are either both } \infty \text{ or both } -\infty.$$

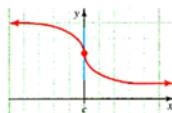
- The graph of f has a **cusp** at P if

$$\lim_{x \rightarrow c^-} f'(x) \text{ and } \lim_{x \rightarrow c^+} f'(x) \text{ are either both infinite with opposite signs.}$$



a. Vertical tangent

$$\begin{aligned} \lim_{x \rightarrow c^-} f'(x) &= \infty \\ \lim_{x \rightarrow c^+} f'(x) &= \infty \end{aligned}$$



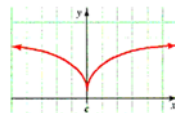
b. Vertical tangent

$$\begin{aligned} \lim_{x \rightarrow c^-} f'(x) &= -\infty \\ \lim_{x \rightarrow c^+} f'(x) &= -\infty \end{aligned}$$



c. Cusp

$$\begin{aligned} \lim_{x \rightarrow c^-} f'(x) &= \infty \\ \lim_{x \rightarrow c^+} f'(x) &= -\infty \end{aligned}$$



d. Cusp

$$\begin{aligned} \lim_{x \rightarrow c^-} f'(x) &= -\infty \\ \lim_{x \rightarrow c^+} f'(x) &= \infty \end{aligned}$$

Example 31

Sketch the graph of functions

$$f(x) = x^{2/3}(2x + 5) \quad \text{and} \quad g(x) = x^{1/3}(x + 4)$$

4.5 L'Hôpital's Rule

INDETERMINATE FORMS:

$$A - B : \infty - \infty \qquad \frac{A}{B} : \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$$AB : 0 \cdot \infty \qquad A^B : 1^\infty, 0^0 \text{ or } \infty^0$$

L'Hôpital's Rule

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} : \frac{0}{0} \text{ or } \frac{\infty}{\infty} \implies \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Example 32

Find the limits

a. $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$

b. $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$

c. $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}}$

d. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$

Example 33

Find the limits

a. $\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x}$

b. $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 1}}{x}$

Indeterminate Products: $0 \cdot \infty$

Writing the product fg as a quotient:

$$fg = \frac{f}{1/g} \quad \text{or} \quad fg = \frac{g}{1/f}$$

Example 34

Find the limits

a. $\lim_{x \rightarrow 0^+} x \ln x$

b. $\lim_{x \rightarrow (\frac{\pi}{2})^-} \left(x - \frac{\pi}{2}\right) \tan x$

Example 35

Find the limits

a. $\lim_{x \rightarrow \infty} x \sin(\pi/x)$

b. $\lim_{x \rightarrow \infty} \sqrt{x} e^{-x/2}$

Indeterminate Differences: $\infty - \infty$

Example 36

Find the limits

a. $\lim_{x \rightarrow (\frac{\pi}{2})^-} (\sec x - \tan x)$

b. $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$

Example 37

Find the limits

a. $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$

b. $\lim_{x \rightarrow \infty} (x - \ln x)$

Indeterminate Powers: 0^0 , ∞^0 or 1^∞

We want to find the limit

$$L = \lim_{x \rightarrow a} [f(x)]^{g(x)} : 0^0, \infty^0 \text{ or } 1^\infty$$

Taking the natural logarithm

$$\ln L = \lim_{x \rightarrow a} \ln [f(x)]^{g(x)} = \lim_{x \rightarrow a} g(x) \ln f(x)$$

Example 38

Calculate the limits

a. $\lim_{x \rightarrow 0^+} x^x$

b. $\lim_{x \rightarrow \infty} x^{1/x}$

c. $\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x}$

Example 39

Calculate the limits

a. $\lim_{x \rightarrow 1^+} x^{1/(1-x)}$

b. $\lim_{x \rightarrow 0^+} (\cos x)^{1/x^2}$

c. $\lim_{x \rightarrow \infty} \left(\frac{2x-3}{2x+5} \right)^{2x+1}$

4.6 Optimization in the Physical Sciences and Engineering

Steps in Solving Optimization Problems

- 1 Understand the Problem
- 2 Establish the functional relationship

$$y = f(x)$$

- 3 Use the methods of Sections 4.1 and 4.3 to find the absolute maximum or minimum value of f .

Example 40

At noon, on a certain day, a truck is 250 mi due east of a car. The truck is traveling west at a constant speed of 25 mi/h, while the car is traveling north at 50 mi/h.

- a. At what rate is the distance between them changing at time t ?
- b. At what time is the distance between the car and the truck neither increasing nor decreasing?
- c. What is the minimal distance between the car and the truck?

Example 41 (Minimizing time of travel)

A dune buggy is on the desert at a point A located 40 km from a point B , which lies on a long, straight road, as shown in Figure 4.49. The driver can travel at 45 km/h on the desert and 75 km/h on the road. The driver will win a prize if she arrives at the finish line at a point D , 50 km from B , in 84 min or less. What route should she travel to minimize the time of travel? Does she win the prize?

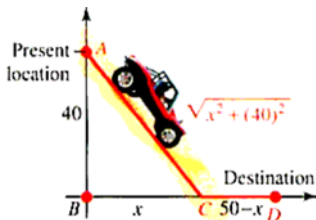


Figure 4.49 A path traveled by a dune buggy

Example 42

A man launches his boat from point A on a bank of a straight river, 3 km wide, and wants to reach point B , 8 km downstream on the opposite bank, as quickly as possible (see Figure 7). He could row his boat directly across the river to point C and then run to B , or he could row directly to B , or he could row to some point D between C and B and then run to B . If he can row 6 km/h and run 8 km/h, where should he land to reach B as soon as possible? (We assume that the speed of the water is negligible compared with the speed at which the man rows.)

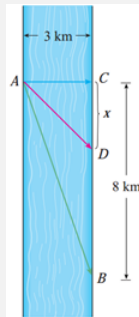


FIGURE 7

Example 43 (Maximizing a constrained area)

You need to build a rectangular fence to enclose a play zone for children. What is the maximum area for this play zone if it is to fit into a right-triangular plot with sides measuring 4 m and 12 m?

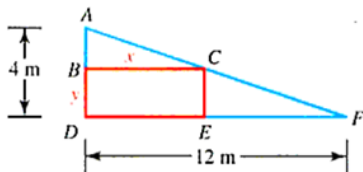
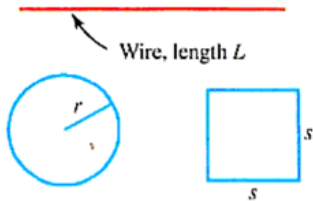


Figure 4.46 Children's play zone

Example 44 (Optimizing a constrained area)

A wire of length L is to be cut into two pieces, one of which will be bent to form a circle and the other to form a square. Determine how the wire should be cut to:

- a. maximize the sum of the areas enclosed by the two pieces.
- b. minimize the sum of the areas enclosed by the two pieces.



Example 45 (Optimizing an angle of observation)

A painting is hung on the wall in such a way that its upper and lower edges are 10 ft and 7 ft above the floor, respectively. An observer whose eyes are 5 ft above the floor stands x feet from the wall, as shown in Figure 4.52. How far away the wall should the observer stand to maximize the angle subtended by the painting?

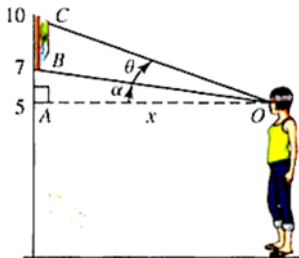


Figure 4.52 Viewing a painting

4.6 Optimization in the Physical Sciences and Engineering

Fermat's Principle of Optics and Snell's Law

PROBLEM

Suppose light travels with speed v_1 in the first medium and speed v_2 in the second medium. What can be said about the path followed by the beam of light?

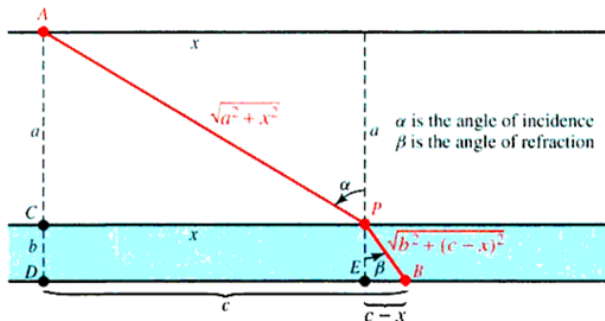


Figure 4.53 The path of a light beam through two media of different density

Fermat's Principle of Optics

Light travels between two points in such a way as to minimize the time of transit.

- Let T_1 and T_2 be the time required for the light to travel from A to P and P to B , respectively. Then the total time of transit is

$$T = T_1 + T_2 = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (c - x)^2}}{v_2}, \quad 0 \leq x \leq c. \quad (2)$$

By Fermat's principle, we want to minimize T as a function of x .

- Find the derivative of T

$$\frac{dT}{dx} = \frac{x}{v_1 \sqrt{a^2 + x^2}} - \frac{c - x}{v_2 \sqrt{b^2 + (c - x)^2}} \quad (3)$$

and noticing that

$$\sin \alpha = \frac{x}{\sqrt{a^2 + x^2}} \quad \text{and} \quad \sin \beta = \frac{c - x}{\sqrt{b^2 + (c - x)^2}}$$

Then we get

$$\frac{dT}{dx} = \frac{\sin \alpha}{v_1} - \frac{\sin \beta}{v_2} \quad (4)$$

- Using the first derivative test, it can be shown that the critical point $P(x, y)$ such that

$$\frac{\sin \alpha}{\sin \beta} = \frac{v_1}{v_2}$$

corresponds to an absolute minimum.

SNELL'S LAW OF FRACTION

If the beam of light strikes the boundary between two media with angle of incident α and is refracted through an angle β , then

$$\frac{\sin \alpha}{\sin \beta} = \frac{v_1}{v_2} =: n \quad (5)$$

where v_1 and v_2 are the rates at which light travels through the first and second medium, respectively. The constant ratio n is called the relative **index of refraction of the two media**.